

Minimum Polyhex Coloring (every color pair shares an edge)

$a(n)$ = minimum cells in a connected polyhex with an n -coloring such that every pair of distinct colors is edge-adjacent.

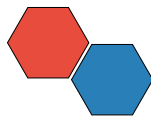
$$a(1..10) = 1, 2, 3, 5, 7, 9, 12, 16, 18, 21$$

Computed by Peter Exley, May 2026

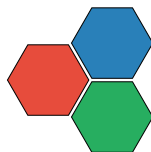
$$a(1) = 1 \text{ [PROVED] } 1 \text{ cells. trivial (single cell); geometric witness verifier}$$



$$a(2) = 2 \text{ [PROVED] } 2 \text{ cells. single-formula SAT; LRAT-certified lower bound; geometric witness verifier}$$



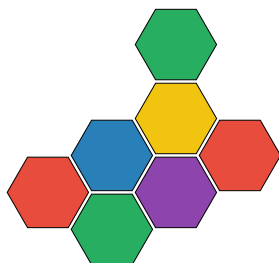
$$a(3) = 3 \text{ [PROVED] } 3 \text{ cells. single-formula SAT; LRAT-certified lower bound; geometric witness verifier}$$



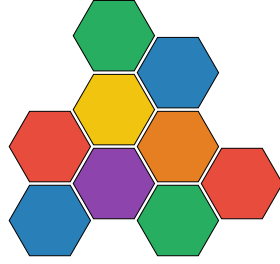
$$a(4) = 5 \text{ [PROVED] } 5 \text{ cells. single-formula SAT; LRAT-certified lower bound; geometric witness verifier}$$



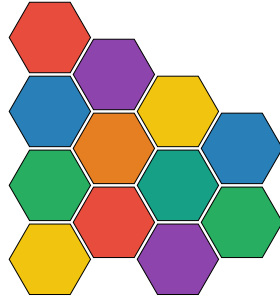
$$a(5) = 7 \text{ [PROVED] } 7 \text{ cells. single-formula SAT; LRAT-certified lower bound; geometric witness verifier}$$



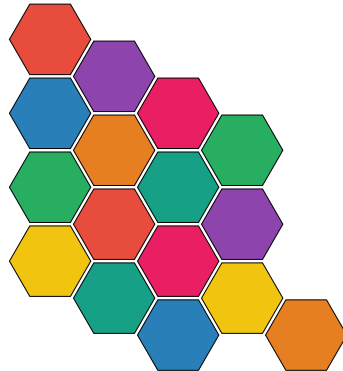
$a(6) = 9$ **[PROVED]** 9 cells. single-formula SAT; LRAT-certified lower bound; geometric witness verifier



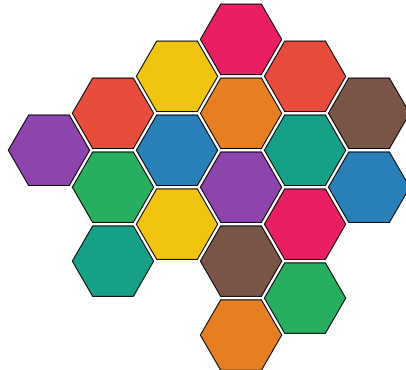
$a(7) = 12$ **[PROVED]** 12 cells. single-formula SAT; LRAT-certified lower bound; geometric witness verifier



$a(8) = 16$ **[PROVED]** 16 cells. rigidity-constrained SAT encoder; analytic lower bound; geometric witness verifier



$a(9) = 18$ **[PROVED]** 18 cells. analytic L2 (per-color) lower bound; geometric witness verifier



$a(10) = 21$ **PROVED** 21 cells. analytic L1 (edge-isoperimetric) lower bound; rigidity-constrained witness; geometric witness verifier

